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Negative Nominal Interest Rates and the Bank Lending Channel

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1 Big picture

- **Question.** Are negative nominal policy rates expansionary once banks hit a deposit lower bound, and how does the bank lending channel transmit negative rates to the macroeconomy?
- **Empirical setting.** The paper studies Sweden, where the Riksbank cut the repo rate below zero several times. The key data advantage is daily bank-level mortgage-rate data combined with aggregate deposit rates, balance-sheet variables, lending volumes, and bank equity prices.
- **Main empirical object.** The authors measure pass-through of repo-rate cuts into bank assets and liabilities: money market rates, government and covered bonds, deposit rates, mortgage lending rates, loan volumes, and bank stock returns.
- **Deposit lower bound (DLB).** Retail household deposit rates stop falling once they are close to zero. In Sweden, pass-through to household deposits weakens as rates go negative and effectively collapses after the repo rate reaches roughly -0.25% .
- **Asset-side pass-through.** Pass-through to money market rates and liquid assets remains strong. This means banks can still earn negative returns on reserves and liquid securities while deposit funding costs stop declining.
- **Lending-rate result.** Before the DLB binds, mortgage-rate pass-through is large. Once the DLB binds, pass-through to mortgage lending rates collapses; average mortgage rates do not fall meaningfully and can even rise after further policy cuts.
- **Bank heterogeneity.** Banks with higher deposit shares have lower pass-through and weaker lending growth under negative rates. Deposit reliance is therefore central to the lending channel.
- **Bank valuation result.** Swedish bank equity prices fall around negative-rate announcements after the DLB binds, suggesting that markets view further negative rates as bad news for bank profitability.
- **Model contribution.** The paper builds a banking model with liquid assets, loans, deposits, external financing, net worth, and a DLB. Negative rates can compress bank profits, reduce net worth, raise lending spreads, and contract output.
- **Sufficient statistic.** Negative rates are expansionary only if the banking sector's net exposure to negative rates is favorable enough. If banks pay negative rates on assets but cannot pass them to deposits, negative rates can be contractionary.
- **Policy takeaway.** The effectiveness of negative policy rates depends on the structure of bank balance sheets and the institutional ability to pass negative rates through to deposits. The zero lower bound is not fully gone; it reappears as a deposit lower bound inside banks.

2 Data and measurement

2.1 Policy-rate environment and Swedish setting

- The Riksbank cut the repo rate into negative territory beginning in 2015.
- Sweden is a useful case because the Riksbank made multiple negative-rate cuts and because the authors observe high-frequency bank-level mortgage rates.
- Sweden is also relatively cash-light, so cash frictions should be less binding than in many countries. This makes the setting a strong test of whether negative rates can work.

Interpretation: The empirical design asks whether the bank lending channel remains effective once the policy rate goes negative and household deposit rates become hard to reduce further.

2.2 Bank balance-sheet variables

The paper organizes the empirical and theoretical analysis around a simplified bank balance sheet:

- **Assets:** loans B , reserves R , money M , and other liquid assets A .
- **Liabilities:** deposits D , external financing F , and net worth N .
- Liquid assets are summarized as $L = M + R + A$.
- The interest rates on these components are denoted i^b , i^r , i^a , i^d , and i^f .

Interpretation: The key friction is that policy rates transmit differently across the two sides of the bank balance sheet. The asset side can go negative even when retail deposit funding cannot.

2.3 Interest-rate data

- The repo rate is the policy rate.
- Money-market rates include STIBOR.
- Liquid-asset rates include Swedish government bonds and covered bonds of different maturities.
- Deposit rates include household and corporate deposit rates.
- Mortgage rates are daily bank-level listed mortgage rates for floating and fixed-rate contracts.

Interpretation: The daily mortgage-rate data are essential because they allow the authors to estimate high-frequency pass-through around policy-rate announcements.

2.4 Deposit lower bound variable

- The DLB is identified empirically as the point at which household deposit-rate pass-through stops responding to additional policy cuts.
- In Sweden, deposit pass-through remains positive for early negative-rate cuts, but the pass-through is essentially zero after the last two cuts.
- The paper’s post-bound period corresponds to the policy cuts to -0.35% and -0.50% .

Interpretation: The lower bound need not bind exactly at zero. It binds when banks become unwilling or unable to reduce retail deposit rates further, which in Sweden occurs after the repo rate reaches roughly -0.25% .

2.5 Lending volumes and bank equity

- Lending growth is measured at the bank level, with attention to deposit shares.
- Bank equity data are daily stock returns for publicly listed Swedish banks.
- Excess bank returns are measured relative to a broad market index around policy-rate announcements.

Interpretation: Lending volumes show whether pass-through failures affect real credit supply. Bank equity responses reveal whether negative rates reduce expected bank profitability.

3 Empirical specifications and results

3.1 Event-study pass-through to mortgage rates

The paper estimates an event-study regression of bank lending rates around policy-rate cuts:

$$\bar{i}_t^l = \alpha + \sum_{k=-30}^{30} \beta_k I_k + I^{DLB} \sum_{k=-30}^{30} \beta_k^{DLB} I_k + \epsilon_t. \quad (1)$$

- $\bar{i}_t^l$ is the average daily aggregate bank lending rate.
- I_k indicates the number of days k since the repo-rate cut.
- I^{DLB} indicates policy cuts after the DLB binds.
- The coefficients β_k capture normal pass-through before the DLB.
- The sum $\beta_k + \beta_k^{DLB}$ captures post-DLB pass-through.

Interpretation: If β_k is negative after a policy cut, mortgage rates fall. If $\beta_k + \beta_k^{DLB}$ is near zero or positive, pass-through has collapsed.

3.2 Deposit-share heterogeneity

The paper relates post-bound pass-through to each bank's deposit share:

$$PassThrough_i = \alpha + \delta DepositShare_i + u_i. \quad (2)$$

Interpretation: Banks that rely more heavily on deposits should suffer more from the DLB, because their marginal funding costs fall less when the policy rate is cut.

3.3 Lending growth specification

The bank-level lending regression is summarized as

$$\Delta \log Loans_{i,t} = \alpha_i + \tau_t + \theta Post_t \times DepositShare_i + \epsilon_{i,t}. \quad (3)$$

- $Post_t$ equals one when the repo rate is below -0.25% .
- The key coefficient is θ .
- A negative θ means that high-deposit-share banks reduce lending growth more after the DLB binds.

Interpretation: This test links pricing pass-through to quantities: banks that cannot lower deposit costs reduce credit growth more.

3.4 Bank equity event study

The paper studies bank stock excess returns around policy announcements:

$$ExcessReturn_{i,t} = \alpha + \gamma_k EventDay_k + \beta_k EventDay_k \times I_t^{post-zero} + \epsilon_{i,t}. \quad (4)$$

Interpretation: Negative β_k coefficients around the announcement day imply that negative-rate cuts after the DLB are viewed as bad news for bank profits.

4 Theoretical framework

4.1 Partial-equilibrium bank problem

Banks choose loans, liquid assets, deposits, external finance, and net worth subject to balance-sheet constraints and costs. The key state variables are:

- liquid-asset holdings and their returns;
- deposit funding and the deposit lower bound;
- external finance and its risk/liquidity cost;
- bank net worth, which affects lending intermediation costs.

Interpretation: The bank is not a frictionless pass-through vehicle. Negative rates affect its funding mix, profitability, net worth, and lending spreads.

4.2 Deposit lower bound in the model

Households value money-like assets up to a satiation point. Once they are satiated, deposit rates cannot be pushed sufficiently negative:

$$S' \left(\frac{M_t^s}{P_t} \right) = \frac{i_t^d}{1 + i_t^d}. \quad (5)$$

Interpretation: The DLB is modeled as the point where household money/deposit demand becomes very elastic. Banks cannot force retail deposit rates down without losing deposits.

4.3 Policy-rate cuts in partial equilibrium

When the DLB is not binding, lower policy rates reduce bank funding costs and borrowing rates. When the DLB binds, lower rates can reduce returns on liquid assets and reserves without reducing deposit costs.

Interpretation: This margin compression lowers bank net worth. Lower net worth raises lending intermediation costs and can push borrowing rates up rather than down.

4.4 General equilibrium and sufficient statistic

The model embeds banks in a New Keynesian environment with borrowers, savers, firms, sticky prices, and a Taylor rule. The key sufficient statistic is the banking sector's net exposure to negative rates, schematically:

$$NetExposure \approx \frac{\bar{L}}{\bar{\Lambda}} + \rho^a \frac{\bar{A}}{\bar{\Lambda}} - \rho^f \frac{\bar{F}}{\bar{\Lambda}}. \quad (6)$$

Interpretation: If the banking sector is exposed to negative rates through assets but cannot transmit them to deposit liabilities, the interest-rate cut multiplier can turn negative.

5 Identification and empirical design

5.1 Why event studies are useful

- Daily mortgage rates allow the paper to isolate rate changes around policy announcements.
- The event-study compares pass-through before and after the DLB binds.
- By scaling mortgage-rate changes by the size of the repo-rate cut, the coefficients have a direct pass-through interpretation.

Interpretation: This design reduces confounding from slow-moving macro variables, but it still depends on the assumption that other announcement-day policies do not dominate the observed rate changes.

5.2 Why deposit-share heterogeneity matters

- Banks differ in their reliance on deposit funding.
- Deposit-heavy banks are more affected by a collapse in deposit-rate pass-through.
- The negative relation between deposit share and mortgage-rate pass-through is therefore a direct test of the bank lending channel.

Interpretation: Deposit-share heterogeneity connects the DLB to bank-specific pricing and quantity responses.

5.3 External validity

- Sweden's banking system is highly developed and relatively cash-light.
- The findings may be stronger or weaker elsewhere depending on deposit spreads, competition, bank funding structures, and household cash usage.
- The model provides a way to translate the Swedish evidence into a broader sufficient-statistic framework.

Interpretation: The empirical estimates are Swedish, but the mechanism is general: negative rates are most problematic when deposits are an important nonnegative-rate liability.

6 Tables, figures, and equation panels

6.1 Figure 1: Bank balance sheet

Read:

- The balance sheet separates assets into loans, reserves, money, and other liquid assets.
- Liabilities are deposits, external financing, and net worth.
- This structure organizes both the empirical evidence and the theoretical model.

Economic message: The DLB matters because different components of the balance sheet have different pass-through to negative policy rates.

	Assets	Liabilities
	Loans (B) -- i^b	Deposits (D) -- i^d
	Reserves (R) -- i^r	External funding (F) -- i^f
	Money (M) -- 0	
L	Other liquid Assets (A) -- i^a	Net worth (N)

FIGURE 1
Bank balance sheet

6.2 Figures 2–4: Pass-through to liquid assets, deposits, and funding costs

Read:

- Figure 2 shows strong pass-through from the repo rate to money market rates and liquid assets.
- Figure 3 shows that deposit-rate pass-through collapses once the repo rate becomes sufficiently negative.
- Figure 4 shows that average bank funding costs stop falling in the same way once deposits hit the lower bound.

Economic message: Negative rates still affect bank assets, but they stop reducing a major liability cost. This creates the bank profitability squeeze.

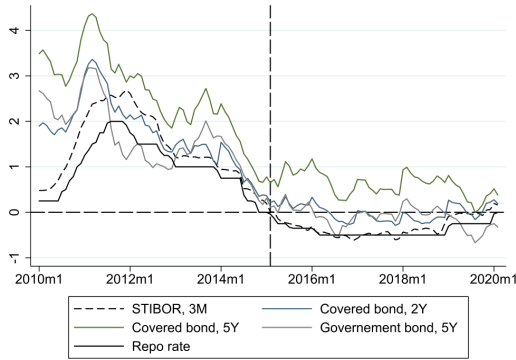


FIGURE 2
Interest on liquid assets

Figure 2 shows the repo-rate together with the interest rates on 1-week interbank rate (STIBOR, 1 week), 1 month government certificate, 1 month, and 2 year government bonds (Government bond, 2 year). Source: Riksbank (2022).

(a) Figure 2: Interest on liquid assets

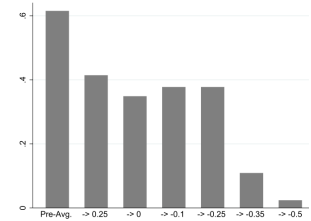
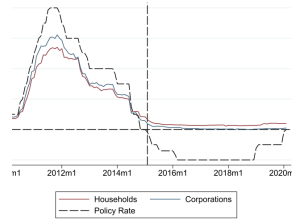


FIGURE 3
Deposit rates (left panel) and the relative change in deposit rates (right panel)

Figure 3 shows the policy rate, household deposit rate, and corporate deposit rate (left panel) and the pass-through of policy rate changes (right panel). The policy rate is defined as the repo rate. In the right panel, we show the change in the deposit rate to the change in the policy rate (i.e. $\Delta i_d / \Delta i_r$) for the last six policy rate reductions. The numbers on the x-axis denote the policy rate reduction. Source: SCB (2022).

(b) Figure 3: Deposit rates and pass-through

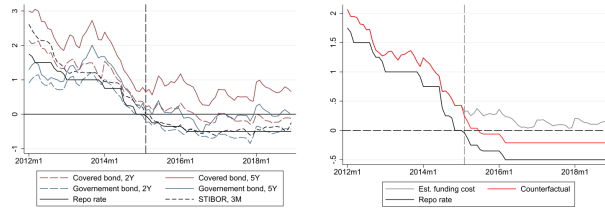


FIGURE 4

Other interest rates (left panel) and an estimate of average bank funding costs (right panel). The estimated average funding cost is computed by taking the weighted average of the assumed interest rates of the different funding sources of the bank. Certificates are assumed to have the same interest rate as 2Y-covered bonds, while unsecured debt is assumed to have the same interest rate as 2Y-covered bonds plus a 2% constant risk-premium. The counterfactual series corresponds to the case when the spread between the repo-rate and the estimated funding cost remains fixed at pre-negative levels. Weights are based on the liability structure of Swedish banks, see Figure A.1. Source: Riksbank (2022).

6.3 Figures 5–6 and Table 1: Mortgage-rate pass-through

Read:

- Figure 5 shows bank-level mortgage rates around the negative-rate period for maturities of 3 months, 1 year, 3 years, and 5 years.
- Table 1 reports the event-study estimates. Before the DLB, pass-through reaches roughly 78% within 30 days.
- The DLB interaction offsets the normal pass-through. At 30 days, the post-bound pass-through is about 5.8%, effectively zero.
- Figure 6 visualizes the collapse: the estimated pass-through after the DLB is far below the normal pass-through path.

Economic message: The central empirical result is that negative policy cuts stop reducing mortgage lending rates once deposit rates are bounded.

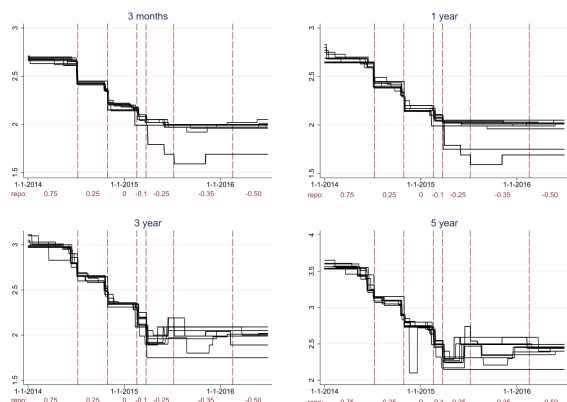


FIGURE 5
Bank-level mortgage rates for different maturities

Notes: This figure shows bank-level mortgage rates for different maturities—3 months, 1, 3, and 5 years, 1 January 2014 to 31 December 2018. Dashed vertical lines indicate days of repo rate reductions. Source: [Comptroller \(2019\)](#).

(a) Figure 5: Bank-level mortgage rates

TABLE 1
Regression results from estimating Equation (2)

Event time	β_k	β_k^{DLB}	Pass-through post-bound: $\beta_k + \beta_k^{DLB}$
-30	0.00	0.00	0.00
-25	-2.94	3.32	0.38
-20	-0.45	8.93	8.48
-15	0.06	8.43	8.48
-10	-4.89	13.6	8.72
-5	-23.2 [*]	31.5	8.33
0	-40.3 ^{***}	48.7 [*]	8.33
5	-62.4 ^{***}	68.3 ^{***}	5.92
10	-72.4 ^{***}	78.4 ^{***}	5.92
15	-73.6 ^{***}	79.5 ^{***}	5.92
20	-67.4 ^{***}	73.3 ^{***}	5.92
25	-71.4 ^{***}	77.3 ^{***}	5.92
30	-77.6 ^{***}	83.4 ^{***}	5.82

(b) Table 1: Event-study estimates

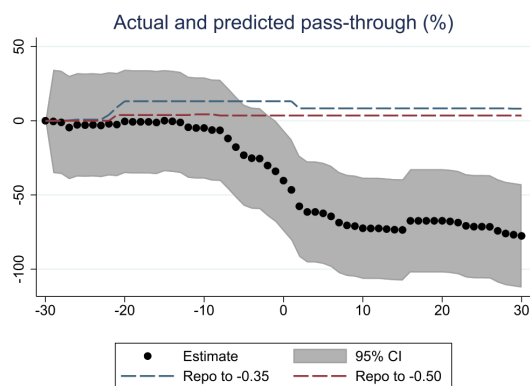


FIGURE 6

Event study on bank-level mortgage rates

Figure 6 compares the actual and predicted pass-through for the two last policy rate cuts in our sample (repo rate reduced to -0.5). The interest rates are weighted average of 3 months, 1, 3, and 5 year fixed interest rate. The repo rate cuts in the period 2009–2015m4 are used to estimate the average pass-through and the corresponding predicted pass-through.

6.4 Table 2 and Figure 7: Deposit shares and pass-through

Read:

- Table 2 regresses post-bound pass-through on bank deposit shares.
- The continuous specification gives a negative coefficient of about -0.957 .
- Figure 7 shows the same negative relation graphically: high-deposit-share banks have lower post-bound pass-through.

Economic message: Deposit reliance is the balance-sheet variable that converts the DLB into a lending-rate pass-through failure.

	(1) Pass-through	(2) Pass-through
Deposit share	-0.957** (-2.15)	-12.49 (-1.68)
Deposit share variable:	continuous	high or low
N	18	18

Notes: This table shows regression results from estimating Equation (3). t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

(a) Table 2: Pass-through and deposit shares

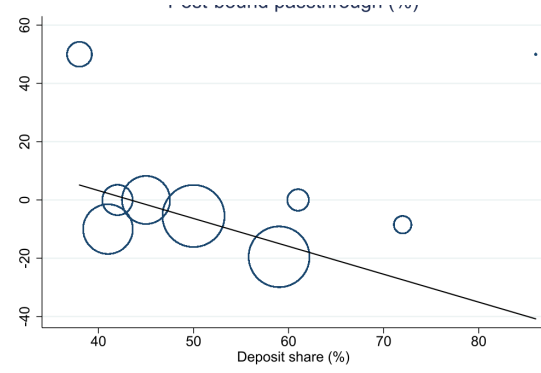


FIGURE 7
Pass-through and deposit shares
The figure shows the average pass-through in the post-bound period as a function of 2014 deposit shares. The size of the circles represents the number of banks in each deposit share bin. The line is a linear regression of bank-level pass-through on deposit share, where each bank is weighted equally.

(b) Figure 7: Deposit share and pass-through

6.5 Table 3 and Figure 8: Lending quantities and bank equity

Read:

- Table 3 shows that high-deposit-share banks have lower lending growth after the DLB binds.
- In the continuous specification, increasing the deposit share from 50% to 60% reduces lending growth by about 2.2 percentage points.
- Figure 8 shows negative bank stock excess returns around negative-rate cuts in the post-zero period, especially around the announcement day.

Economic message: The DLB affects quantities and profitability, not only rates. Negative cuts are interpreted by markets as adverse for bank net worth.

	(1) $\Delta \log(\text{loans})$	(2) $\Delta \log(\text{loans})$
Post $\times$ deposit share	-0.217** (0.096)	-3.235** (1.185)
N	361	361
No. of clusters	10	10
Mean of dependent variable	1.932	1.932
SD of dependent variable	9.113	9.113
Bank FE	Yes	Yes
Time FE	Yes	Yes
Measure	Cont.	High or low

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Post = 1 for the period when the repo-rate is below -0.25, and zero otherwise. Standard errors clustered at the bank level. Summary statistics taken over whole sample period (2014.1–17).

(a) Table 3: Bank lending growth

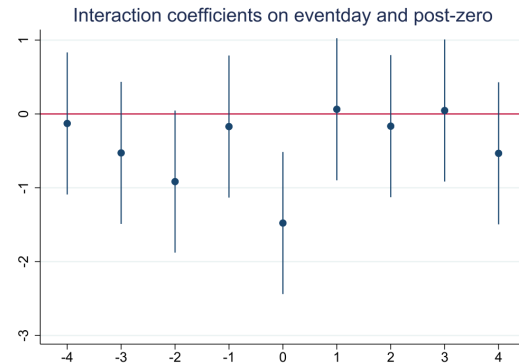


FIGURE 8
Excess stock return and negative policy rates
The figure shows coefficient estimates $\hat{\beta}_k$ from estimating Equation (5). The coefficient estimates can be interpreted as the excess return of bank stocks in the post-zero period in a window around policy rate cut announcement days (k days before/after). Data from OMX (2023).

(b) Figure 8: Bank excess returns

6.6 Equation panels: Event study, DLB, and partial equilibrium

Read:

- The event-study equation defines normal and DLB-specific pass-through paths.

- The deposit-bound equations describe how money/deposit demand generates a lower bound on deposit rates.
- The partial-equilibrium equations show how borrowing, net worth, and policy-rate pass-through are linked.

Economic message: These equations are the bridge from empirical pass-through to the model's sufficient-statistic logic.

ay demand is given by:

$$\frac{m' \left(\frac{M_t^s}{P_t} \right)}{U'(C_t^s)} = \frac{i_t^d}{1 + i_t^d} - S' \left(\frac{M_t^s}{P_t} \right) \quad (24)$$

if $m_t \geq \bar{m}$ the household is satiated in money and

$$\bar{i}_t^l = \alpha + \sum_{k=-30}^{30} \beta_k I_k + I_t^{\text{DLB}} \sum_{k=-30}^{30} \beta_k^{\text{DLB}} I_k + \epsilon_t \quad (2) \quad S' \left(\frac{M_t^s}{P_t} \right) = \frac{i_t^d}{1 + i_t^d} \quad (25)$$

is the average, daily aggregate bank lending rate. We construct this variable by aggregating

tion (25) generates the bound on deposit rates.

(a) Event-study equation

(b) Deposit lower bound conditions

log-linear approximation of (11) and (13) around the steady-state yields

$$\hat{B}_t = \frac{1}{v-1} \left(\frac{1 + \bar{i}^b}{\bar{i}^b - \bar{i}^d} \right) (\hat{i}_t^b - \hat{i}_t^d) + \frac{1}{v-1} \hat{N}_t \quad (16)$$

$$\hat{N}_t = \frac{1 + \bar{i}^d}{\Theta} \left\{ (1 - \omega) \hat{N}_{t-1} + \frac{\bar{B}}{\bar{N}} \bar{i}_t^b \frac{1 + \bar{i}^b}{1 + \bar{i}^d} + \Omega_\rho \hat{r}_t^r \right\} \quad (17)$$

where $\hat{N}_t \equiv \log N_t / \log \bar{N}$, $\hat{B}_t \equiv \log B_t / \log \bar{B}$, $\hat{i}_t^d \equiv \log(1 + i_t^d) / (1 + \bar{i}^d)$, $\hat{i}_t^b \equiv \log(1 + i_t^b) / (1 + \bar{i}^b)$, and $\Theta \equiv 1 - \frac{1}{v} (\bar{i}^b - \bar{i}^d) \frac{\bar{B}}{\bar{N}} > 0$. The key coefficient is Ω_ρ , defined and discussed in detail below.

In the remaining analysis, we make the following assumption

$$\frac{1}{v} < \frac{(1 - \omega)(1 + \bar{i}^d) \bar{N}}{\bar{i}^b - \bar{i}^d \bar{B}} \quad (18)$$

Equation (18) says that the feedback effect from changes to net worth to lending is not too strong, which is a necessary condition for the approximate solution to have a unique bounded solution. Observe

6.7 Tables 4–6: Calibration and model moments

Read:

- Table 4 compares changes in external financing, liquid assets, and deposits per 1 percentage point policy-rate cut when the repo rate is negative.
- The model matches the data: external financing falls slightly, liquid assets fall substantially, and deposits rise.
- Table 5 sets banking-sector moments: liquid assets are 42% of total assets, reserves 5%, external financing 53%, loans 53%, and net worth 5%.
- Table 6 calibrates structural parameters, including the marginal cost of lending, the elasticity of lending costs with respect to net worth, the payout ratio, and the preference shock.

Economic message: The calibration is disciplined by Swedish banking data and by the pass-through estimates; it is not an arbitrary model exercise.

TABLE 4
Change in external financing, liquid assets, and deposits per 1 percentage point reduction in i_t^r when $i_t^r < 0$: data versus model

Moment	Data	Model
$\Delta \frac{E_t}{N_t}$	-0.2%	-0.3%
$\Delta \frac{L_t}{N_t}$	-7.5%	-6.8%
$\Delta \frac{D_t}{N_t}$	1.6%	1.9%

(a) Table 4: Data versus model moments

TABLE 5
Banking sector parameters

Banking Sector Moments	Value	Empirical Counterpart
Liquid asset ratio	$\frac{\bar{L}}{\bar{A}} = 0.42$	Liquid assets to total assets of 42%
Reserve ratio	$\frac{\bar{R}}{\bar{A}} = 0.05$	Reserves to total assets of 5%
External financing ratio	$\frac{\bar{F}}{\bar{A}} = 0.53$	External financing to total assets of 53%
Loans to total assets	$\frac{\bar{B}}{\bar{A}} = 0.53$	Loans to total assets of 53%
Net worth to total assets	$\frac{\bar{N}}{\bar{A}} = 0.05$	Net worth to total assets of 5%
Profit to net worth	$\frac{\bar{\pi}}{\bar{N}} = 0.173$	Profit to net worth of 17.3%
Loans to GDP	$\frac{\bar{B}}{\bar{Y}} = 1.4$	Loans to GDP of 140%
Pass-through coefficient to liquid assets	$\rho^a = 1$	Pass-through of negative rates of 100%
Pass-through coefficient to external financing	$\rho^f = 0.4$	Pass-through of external financing of 40%

Notes: This table shows key banking sector moments and pass-through coefficients necessary to calibrate the log-linear approximated model outlined in [Appendix D](#). Banking sector moments are based on the 2014 balance sheet of the

(b) Table 5: Banking-sector parameters

TABLE 6
Calibration

Parameter	Value	Source/Target
Intertemporal elasticity of substitution	$\sigma = 0.66$	Smets and Wouters (2003)
Share of borrowers	$\chi = 0.61$	Justiniano et al. (2015)
Steady-state gross inflation rate	$\bar{\pi} = 1.005$	Match annual inflation target of 2%
Discount factor, saver	$\beta^s = 0.9963$	Annual real savings rate of 1.5%
Discount factor, borrower	$\beta^b = 0.991$	Annual real borrowing rate of 3.5% (Mehra and Prescott, 2008)
Slope of AS equation	$\kappa = 0.02$	Eggertsson and Woodford (2003)
Taylor coefficient on inflation gap	$\phi_\pi = 1.5$	Gali (2008)
Taylor coefficient on output gap	$\phi_Y = 0.5/4$	Gali (2008)
Marginal cost of lending	$\nu = 4.5$	Match pass-through of negative rates from Table 1.
Elasticity of lending costs wrt. net worth	$\iota = 2.28$	$\iota = (\nu - 1) \times 1.89/2.89$, based on MAG (2010) and model equations
Payout ratio	$\omega = 0.17$	Generate profit to net worth ratio of 17.3%
Shock	Value	Source/Target
Preference shock	22.7% temporary decrease in c_t	Generate a 4.5 drop in output on impact
Persistence of preference shock	$\rho = 0.92$	Duration of lower bound of 12 quarters

6.8 Figures 9–11 and Table 7: General equilibrium effects

Read:

- Figure 9 compares impulse responses under three regimes: negative rates with DLB, a zero lower bound on the policy rate, and no lower bounds.
- Under the calibrated DLB case, negative policy cuts deepen the output contraction because deposit rates do not fall while bank profits and net worth decline.
- Figure 10 decomposes the mechanism: deposit rates are stuck, borrowing rates rise relative to the frictionless benchmark, and bank net worth falls.
- Figure 11 shows that the effect of negative rates depends on banks' net exposure to negative rates.
- Table 7 reports output interest-cut multipliers under different deposit-market assumptions; the baseline multiplier is negative, and alternative assumptions can make it more contractionary.

Economic message: Negative rates are not always expansionary. They work only when the banking system is positioned so that policy cuts reduce borrowing rates rather than compress bank net worth.

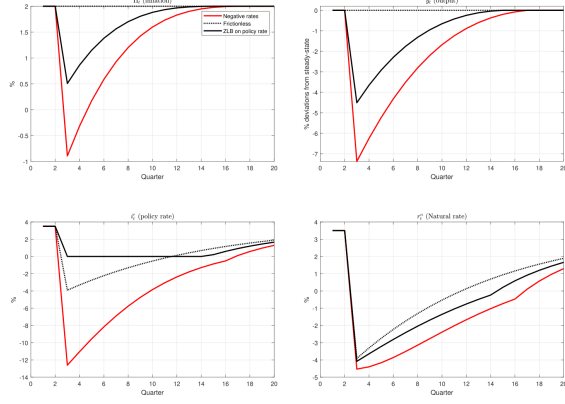


FIGURE 9
Response of model under the baseline calibration to a preference shock

Notes: This figure shows the impulse response functions of inflation, output, the policy rate and the natural rate of interest in response to a preference shock. The red solid line corresponds to a model where the central bank follows the Taylor rule into negative territory, but in which the deposit rate is subject to a lower bound. The black solid line corresponds to a model where the central bank follows the Taylor rule in positive territory, but where the policy rate is subject to a ZLB. The dashed black line corresponds to a model in which there are no bounds on any interest rate.

calibrated to match the average decline in real output in Switzerland, the Euro Area, Sweden

(a) Figure 9: Baseline impulse responses

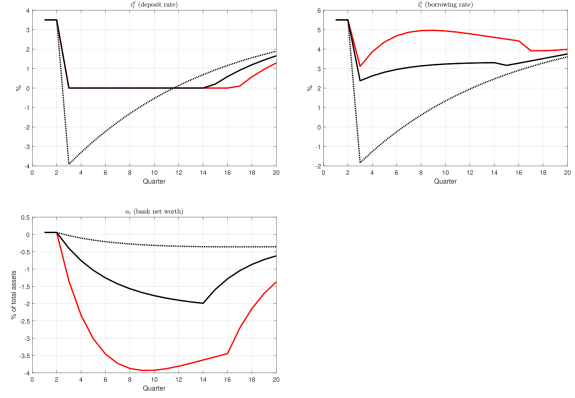


FIGURE 10
Response of model under the baseline calibration to a preference shock

Notes: This figure shows the impulse response functions of the deposit rate, borrowing rate, and bank net worth, in response to a preference shock. The red solid line corresponds to a model where the central bank follows the Taylor rule into negative territory, but in which the

(b) Figure 10: Mechanism

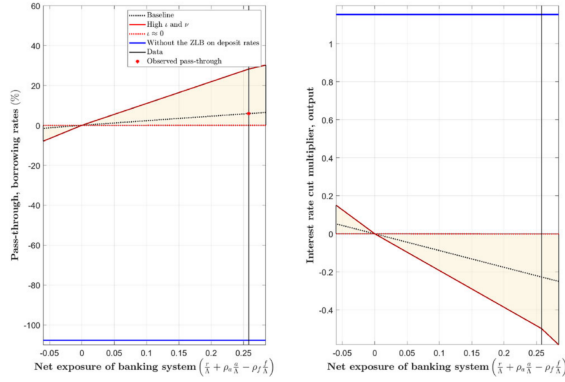


FIGURE 11

Effect of negative policy rates and net exposure of the banking system

Notes: This figure shows the on-impact difference in borrowing rates (left) and output (right) between two models—one model where the central bank does not impose a negative policy rate and one where the central bank follows a standard Taylor rule also below zero. The output and borrowing rate difference is scaled by absolute value of the average policy rate below zero. On the x-axis, we redo the exercise for different values of Equation (46). We fix R , A , and F and solve the model for different values of $\rho^f \in (0.3, 1)$. The black dashed line is the results using our baseline calibration. The red solid line is a calibration where $v = 51.5$, which generates a pass-through at the DLB equal to the coefficient estimate in Table 1 plus 1 standard deviation. The blue horizontal lines show the effects in the case when the DLB

(a) Figure 11: Sufficient statistic

TABLE 7
Interest cut multiplier (output), with different assumptions about deposit frictions

Model of deposit market	Interest cut multiplier, output
Perfect competition (baseline)	-0.227
Adjustment costs (Appendix D.9.1)	-0.242
Monopolistic competition (Appendix D.9.2)	-0.492

which external financing barely changed while deposits increased. Since net worth declines as a

(b) Table 7: Deposit-market assumptions

7 Section-by-section summary

7.1 Section 1: Introduction

- Negative policy rates challenge the traditional zero lower bound view, but their effectiveness depends on how they transmit through banks.
- The paper argues that retail deposits introduce a new lower bound inside the banking system.
- The main contribution is to combine Swedish microdata with a general equilibrium model calibrated to the empirical pass-through evidence.

7.2 Section 2: Empirical analysis

- Liquid-asset rates respond strongly to negative rates.

- Deposit rates do not keep responding once they approach the DLB.
- Mortgage-rate pass-through collapses after the DLB binds.
- Deposit-heavy banks have weaker pass-through and lower lending growth.
- Bank equity prices decline around post-DLB negative-rate announcements.

7.3 Section 3: Theoretical analysis

- The model builds a bank balance sheet with loans, liquid assets, deposits, external finance, and net worth.
- A DLB arises because households do not accept sufficiently negative deposit rates.
- Negative policy rates can reduce bank returns on liquid assets and reserves without lowering deposit costs.
- Lower bank net worth raises intermediation costs and borrowing rates.
- The model delivers a sufficient statistic based on bank balance-sheet exposure.

7.4 Section 4: Making negative rates more effective

- Negative rates are more effective when deposit-rate pass-through is larger.
- They are also more effective when banks have less unfavorable net exposure to negative rates.
- Institutional arrangements that weaken the DLB or reduce bank profitability compression can improve the effectiveness of negative policy rates.

7.5 Section 5: Conclusion

- The paper concludes that negative nominal rates may be contractionary through the bank lending channel once the DLB binds.
- The effectiveness of negative rates is an empirical and institutional question, not a simple consequence of moving the policy rate below zero.

8 Short summary

- Sweden's negative-rate episode provides high-frequency evidence on the bank lending channel.
- Money-market and liquid-asset rates follow the policy rate into negative territory.
- Household deposit rates hit a lower bound and stop falling.
- Mortgage-rate pass-through collapses once the DLB binds.
- Deposit-heavy banks have lower pass-through and lower lending growth.
- Bank stock prices fall around post-DLB negative-rate announcements.
- The model explains these facts through bank profitability and net worth.
- Negative rates are expansionary only if they lower borrowing rates rather than compress bank balance sheets.

9 Conclusion

9.1 Core conclusion

Eggertsson, Juelsrud, Summers, and Wold show that negative nominal interest rates do not automatically relax monetary policy constraints. Once household deposits hit a lower bound, further policy-rate cuts may fail to reduce lending rates and may instead reduce bank profitability and lending.

9.2 Economic intuition

Banks fund themselves heavily with deposits. If deposit rates cannot fall, a policy-rate cut reduces returns on reserves and liquid assets without reducing a major funding cost. The resulting margin compression lowers bank net worth. Because lending is costly when net worth is low, borrowing rates can rise and credit can contract.

9.3 Takeaway

The zero lower bound reappears inside the banking sector as a deposit lower bound. Negative rates can be useful, but only when the banking system and deposit market allow pass-through. The sufficient statistic is therefore not simply the level of the policy rate, but banks' balance-sheet exposure to negative rates.